PhD DOCTORAL COURSE

Deep Learning for Scientific Discovery

Deep learning as a research instrument across the exact and natural sciences — taught in English.

For doctoral students across the exact and natural sciences — physics, chemistry, mathematics, computer science, and biophysics —
no prior deep-learning experience required.

Every scientific revolution arrives with a new instrument — the telescope, the microscope, the accelerator. Machine learning is the instrument of our era: the same core ideas now fold proteins, read genomes, discover new algorithms, and power the large language models behind Claude and ChatGPT. This course is for doctoral researchers who do not want a black box — you will learn **how deep learning actually works, why it works, and where it fails**, and put it to work on concrete problems across the natural and exact sciences. We build every method **from first principles**, and a few shared principles connect them all, so that once you grasp them you can adapt the same tools to problems in your own field — from everyday data to quantum systems. Mastering them lets you **see what classical methods cannot** and **do original research** anchored in concrete, reproducible science. No prior deep-learning experience is assumed — you leave able to **specify, train, and diagnose your own models**.

Open across the Doctoral School of Exact and Natural Sciences: Physics, Chemistry, Mathematics, Computer Science, Technical Computer Science, and Biophysics.

30 h Lectures	30 h Laboratories	From Data to Discovery
Foundations & theory	Hands-on coding in Python	Worked examples throughout

What you will learn

★ **Understand the AI behind the headlines** — the ideas driving today's landmark breakthroughs, from the **large language models behind Claude and ChatGPT** to the **Nobel-winning AlphaFold** and **AlphaGenome** in the life sciences, and **AlphaTensor** and **AlphaCode** in mathematics and computing — gaining the technical grounding to understand how such systems actually work.

- **How neural networks really work** — build, train, and troubleshoot your own models from the ground up, with no black boxes.
- **How large language models work** — the transformer and self-attention architecture behind models such as Claude and ChatGPT.
- **Reinforcement learning** — how machines learn from reward, and the ideas behind AlphaGo and AlphaZero.
- **Generative AI, from the inside** — the ideas behind models that learn to create, compress, and reimagine data (autoencoders, GANs, diffusion models, and more).
- **Learning dynamics from data** — teach models how systems change over time, including physics-aware models that respect known laws and symmetries.
- **Discovering hidden structure** — find patterns and hidden regularities in data, even when no one tells the model what to look for.
- **The safety of AI** — how neural networks can be fooled and attacked, and how to make them robust.
- **A first look at quantum machine learning** — a short, self-contained glimpse of where deep learning meets quantum computing; no physics background assumed, and a bridge to the follow-up course.

Prerequisites — no deep-learning background needed

- ✓ **Linear algebra & calculus** — matrices, eigenproblems, gradients *(essential)*.
- ✓ **Probability & statistics** — distributions, expectation, sampling.
- ✓ **Python & NumPy** — required; a deep-learning framework (JAX, PyTorch, or Keras) a plus, not required.
- ✓ **Curiosity from any field** — no prior machine-learning background needed.

Dates	Location	Credits
Winter semester 2026/2027	Faculty of Physics, Astronomy & Applied Computer Science, UJ	6 ECTS

Make deep learning a tool for your own research — and know when, and how, to apply neural networks.

Based on the accompanying Springer textbook *Deep Learning in Classical and Quantum Physics* (T. Heightman & M. Płodzień, 2026, arXiv:2508.10666) and the co-authored *Machine Learning in Quantum Sciences* (Dawid et al., Cambridge University Press 2025, arXiv:2204.04198).

Enrolment

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for details and registration